Full Envelope Airspeed Calibration

Introduction

- What is static source position error (SSPE)?
- Kinematic relationships for airspeed calibration
- Traditional methods for airspeed calibration
- The single turning acceleration technique
- Numerical results from C-130J simulations
- Real flight test results

Static Source Position Error (SSPE)

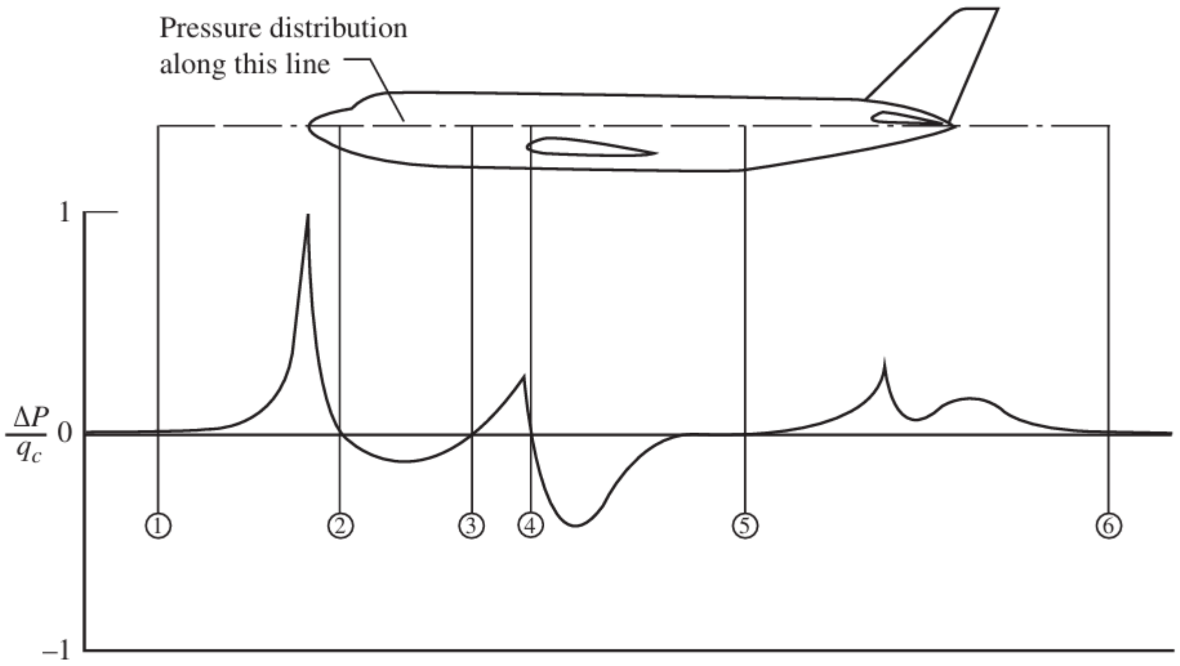

Figure 8.4 The pitot-static probe mounted on an airplane and the surface pressure distribution along the fuselage reference line (FRL). Source: Haering (1995).

- $V_{CAS} = V_{IAS} + \Delta V$
- SSPE is caused by aircraft-induced static pressure distortion
- Airspeed errors are also functions of airspeed (and by corollary, a function of Angle of Attack (AOA) and load factor).

(Gregory & Liu, Introduction to Flight Testing, 2021)

Kinematic Relationships for Airspeed Calibration

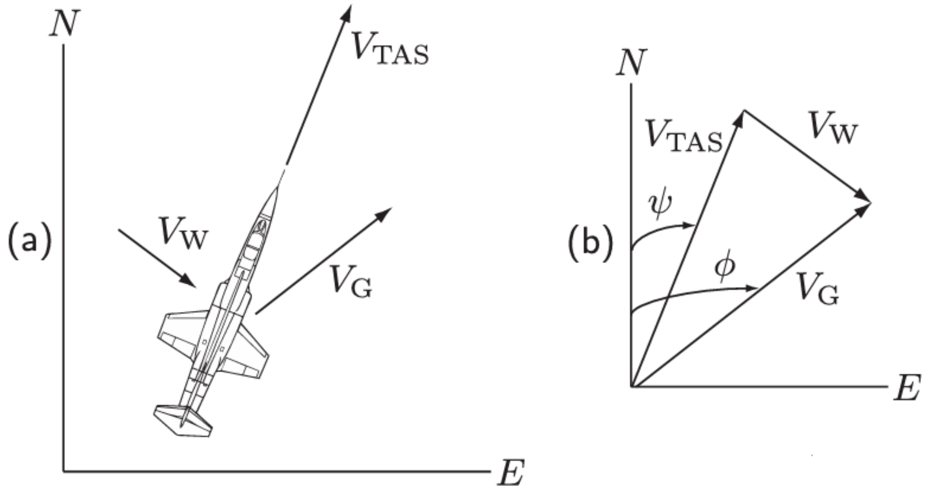

Figure 1: Reference System Definition.

IAS and TAS direction → **heading** ψ

Ground speed direction → **course** ϕ

Wind vector → **unknown**

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Kinematic Relationships for Airspeed Calibration

$$V_{CAS} = a_0 \sqrt{5\left[\left(\frac{q_c}{P_0} + 1\right)^{\frac{2}{7}} - 1\right]}$$

$$V_{TAS} = a_0 \sqrt{5\frac{T}{T_0}\left[\left(\frac{q_c}{P} + 1\right)^{\frac{2}{7}} - 1\right]}$$

$$V_{CAS} = V_{IAS} + \Delta V$$

Assume sea-level flight:

$$T = T_0, \qquad P = P_0$$

$$V_{CAS} = V_{TAS}$$

$$V_{TAS} + V_w = V_G$$

$$\boxed{V_{IAS} + \Delta V + V_w = V_G}$$

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GPS-Based Cloverleaf Calibration Method

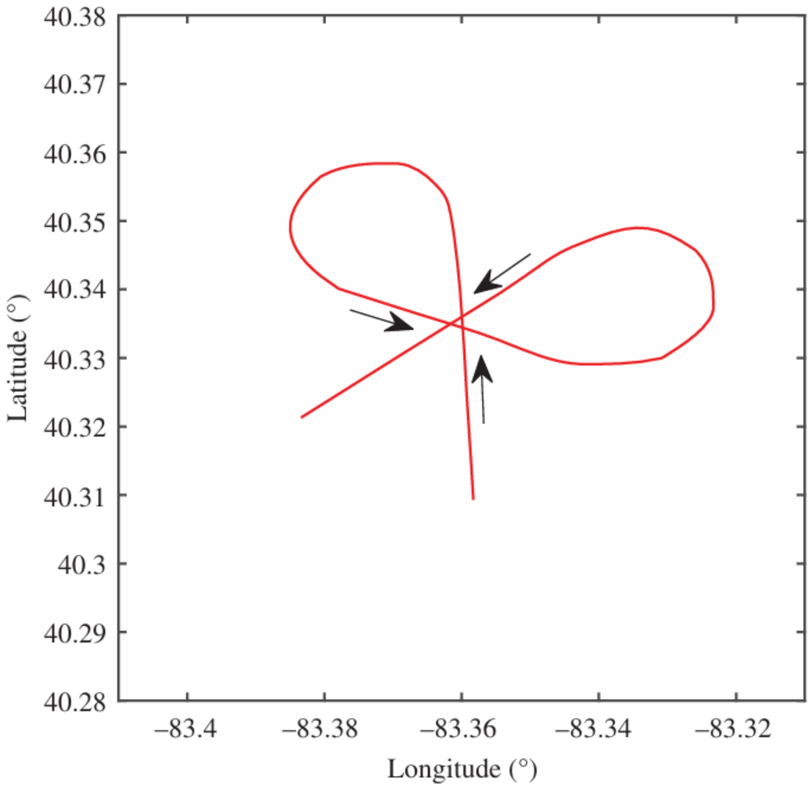

Figure 8.9 Sample ground track (a "cloverleaf") for GPS-based determination of TAS.

Three Unknowns in the Cloverleaf Method:

ΔV
$V_{wind,E}$
$V_{wind,N}$

$$V_{IAS} + \Delta V = V_{TAS} = \sqrt{\left(V_{GS,E} - V_{wind,E}\right)^2 + \left(V_{GS,N} - V_{wind,N}\right)^2}$$

(Gregory & Liu, Introduction to Flight Testing, 2021)

Level Turn Airspeed Calibration Method

North Component

$$V_G \cos\phi = (V_{IAS} + \Delta V_n)\cos\psi + V_{W_N}$$

East Component

$$V_G \sin\phi = (V_{IAS} + \Delta V_n)\sin\psi + V_{W_E}$$

The Matrix Form for Single Point

$$\begin{bmatrix} V_G \cos\phi - V_{IAS}\cos\psi \\ V_G \sin\phi - V_{IAS}\sin\psi \end{bmatrix} = \begin{bmatrix} 1 & 0 & \cos\psi \\ 0 & 1 & \sin\psi \end{bmatrix} \begin{bmatrix} V_{W_N} \\ V_{W_E} \\ \Delta V_n \end{bmatrix}$$

For M = 10^3 Points

$$\begin{bmatrix} V_{G(1)}\cos\phi_{(1)} - V_{IAS(1)}\cos\psi_{(1)} \\ \vdots \\ V_{G(M)}\cos\phi_{(M)} - V_{IAS(M)}\cos\psi_{(M)} \\ V_{G(1)}\sin\phi_{(1)} - V_{IAS(1)}\sin\psi_{(1)} \\ \vdots \\ V_{G(M)}\sin\phi_{(M)} - V_{IAS(M)}\sin\psi_{(M)} \end{bmatrix} = \begin{bmatrix} 1 & 0 & \cos\psi_{(1)} \\ \vdots & \vdots & \vdots \\ 1 & 0 & \cos\psi_{(M)} \\ 0 & 1 & \sin\psi_{(1)} \\ \vdots & \vdots & \vdots \\ 0 & 1 & \sin\psi_{(M)} \end{bmatrix} \begin{bmatrix} V_{W_N} \\ V_{W_E} \\ \Delta V_n \end{bmatrix}$$

- **Overdetermined system**
- No unique solution → **least squares**
- Wind **magnitude and direction are inferred.**

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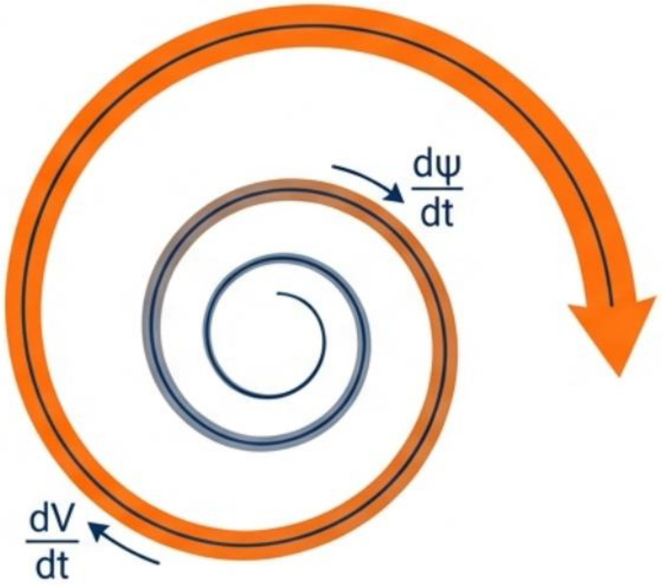
The Single Turning Acceleration Technique



•**Airspeed is intentionally accelerated** during the turn

$$dV/dt \neq 0$$

•**Heading changes continuously**

$$d\psi/dt \neq 0$$

•Each time sample provides an **independent observation** of the wind and airspeed error relationship

•The **static source position error** is modeled as a **function of IAS** rather than a constant bias

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The Single Turning Acceleration Technique

Model Name	Mathematical Form	Type
Polynomial(N)	$\mathcal{M}_{\Delta}(V_{\mathrm{IAS}}; \boldsymbol{w}) = \sum_{i=0}^{P} w_i V_{\mathrm{IAS}}^{i}$	Linear
Power	$\mathcal{M}_{\Delta}(V_{\mathrm{IAS}}; \boldsymbol{w}) = w_0 + w_1 V_{\mathrm{IAS}}^{w_2}$	Nonlinear
Exponential	$\mathcal{M}_{\Delta}(V_{\mathrm{IAS}}; \boldsymbol{w}) = w_0 + w_1 \exp\left(w_2 V_{\mathrm{IAS}}\right)$	Nonlinear

Table 1: Example model families used in fitting SSPE curves.

Several **candidate error model families** are considered:

- Polynomial (flexible, linear in parameters)
- Power-law (smooth monotonic behavior)
- Exponential (rapid nonlinear trends)
- Akaike Information Criterion (AIC/AICc) balances goodness of fit against model complexity.

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The Single Turning Acceleration Technique

Angle of Bank Selection:

- Angle of bank directly determines **turn radius** and **load factor.**
- Increasing AOB increases **load factor**, which in turn increases **angle of attack (AoA)** for a given airspeed.
- Therefore, the **lowest practical angle of bank** is recommended to best approximate wings-level airspeed calibration

Altitude Selection:

- As close as possible to **sea level** or at **higher altitudes** with additional computational and instrumentation requirements.
- The maneuver tolerance should be **±50 feet** to assure a confidence interval of **±0.5 knots.**

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The Single Turning Acceleration Technique

Acceleration Profile:

- At least **one full turn** is required to resolve **wind magnitude and direction**
- A **wide range of airspeeds** is necessary to accurately estimate the SSPE curve
- The acceleration profile should remain **as constant as possible** to minimize wind variability effects
- Separate tests are recommended for **different airspeed regimes** (e.g. subsonic vs transonic)

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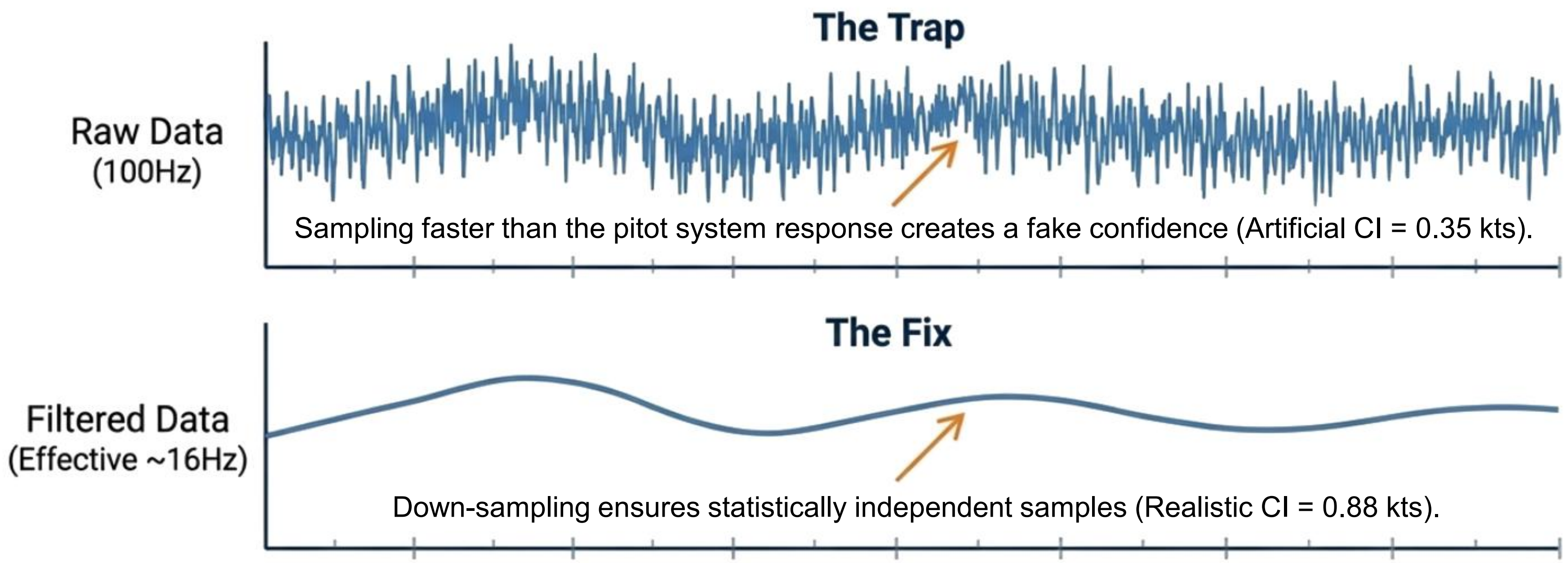
The Single Turning Acceleration Technique

Instrumentation Data Rate



- More data is **not** always better. Sequential correlation in high-speed sensors can artificially shrink confidence intervals.

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Numerical Results from C-130J Simulations

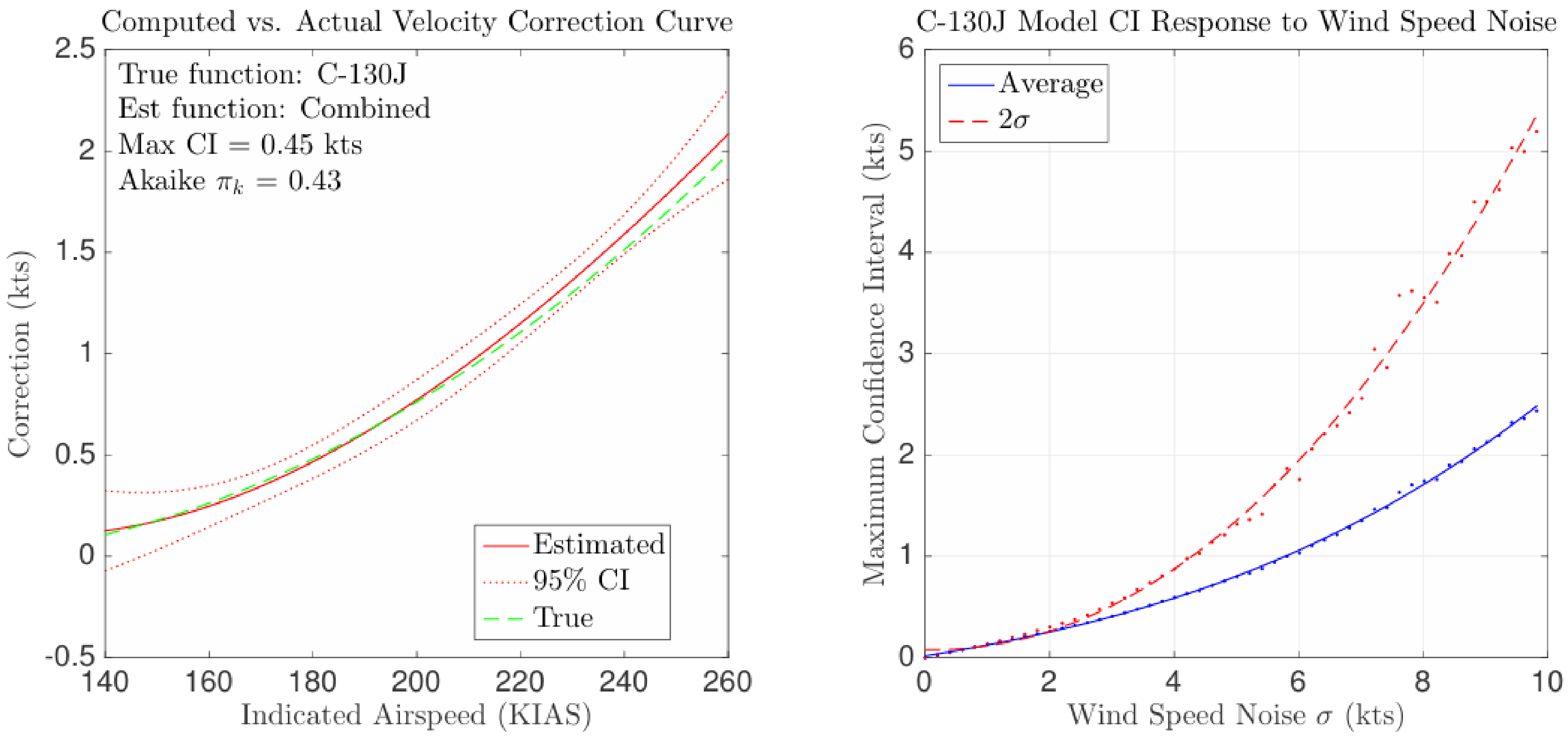

Figure 2: C-130J Numerical Results

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Real Flight Test Results

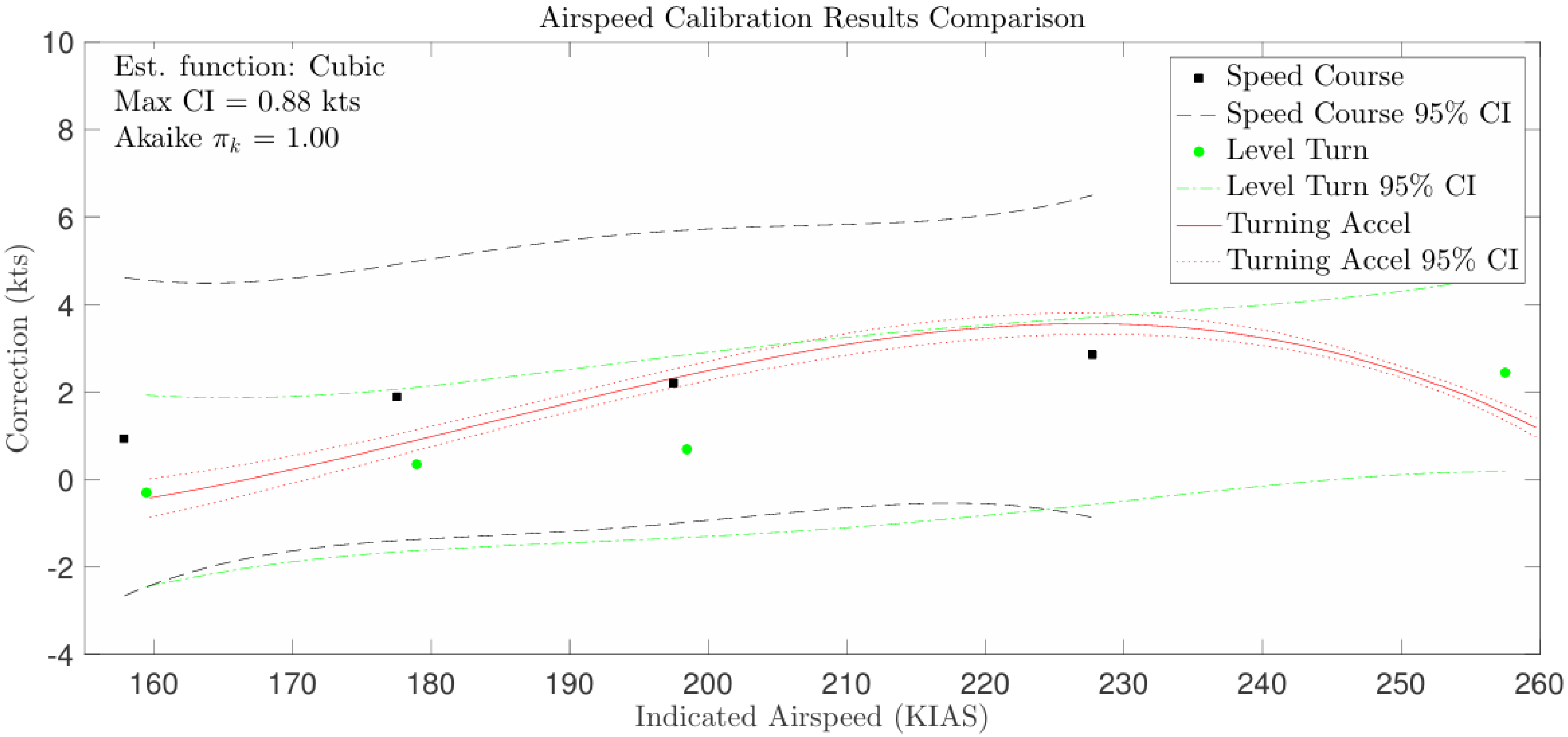

Figure 3: Flight Test Results.

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Real Flight Test Results

FTT	Test Points	Flight Time	Savings
Ground Speed Course	4	36.5 min.	-
Level Turn	4	15.7 min.	57%
Turning Acceleration	1	4.6 min.	87%

Table 2: Flight test time comparison.

In **level turn methods**, gust-affected data points are relatively easy to identify.

- Measurements showing sudden deviations are **discarded** from the analysis
- The results are more accurate for a single **IAS value** (Assuming no drift in wind).
- But flying the whole envelope results in **higher** costs.
- Less tolerant of execution error due to longer flight times.

In the **turning acceleration technique**, airspeed is intentionally varied.

- Gust effects are treated as **disturbances.**
- Their influence is **reduced numerically** through regression and averaging
- Data are **down-weighted**, not discarded
- Because turning acceleration method is quick to execute, it tends to be less affected by wind drifts (not gusts).

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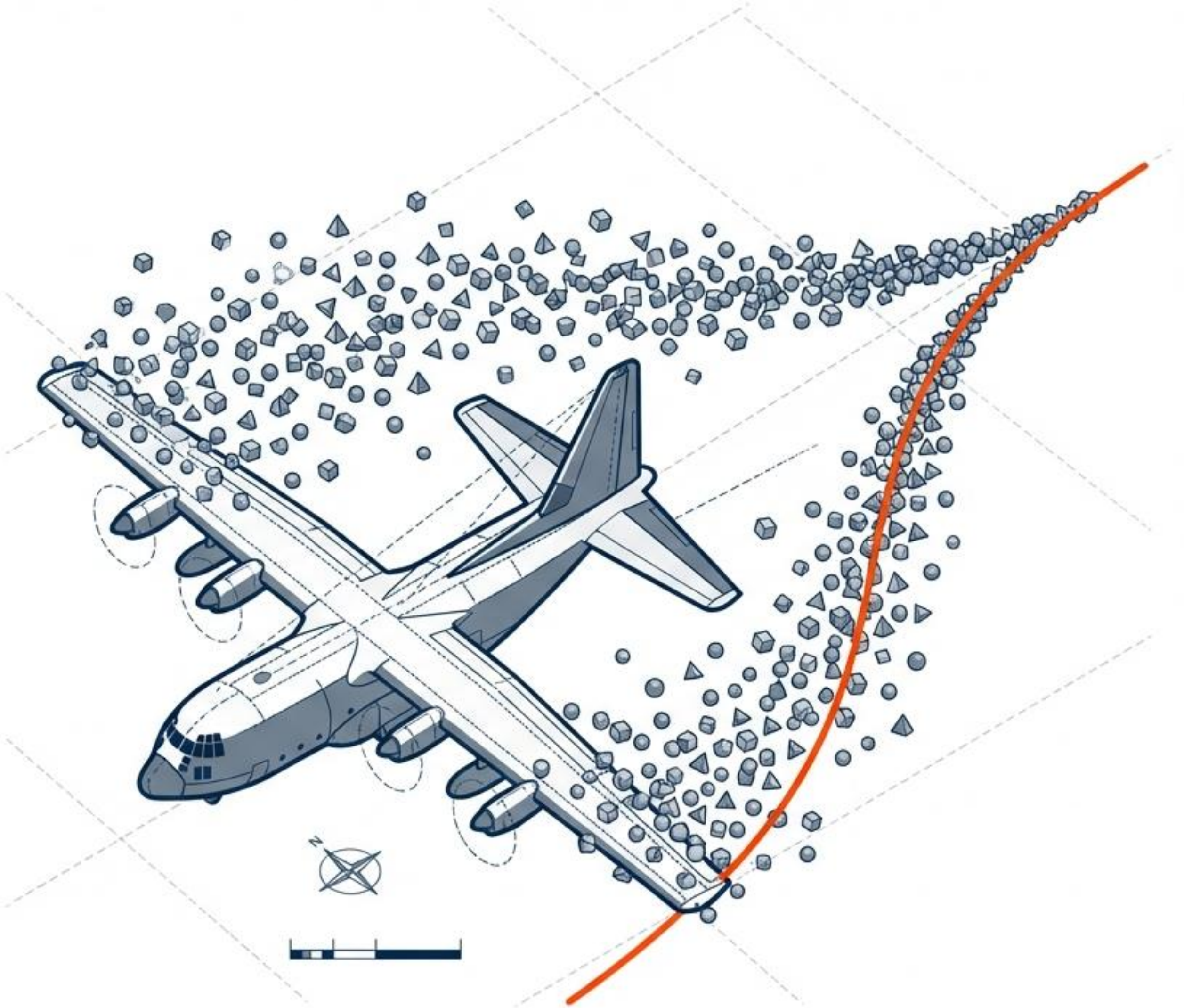

Thank you.

References

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[2] I. M. Gregory and Y. Liu, Introduction to Flight Testing. Wiley, 2021.

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